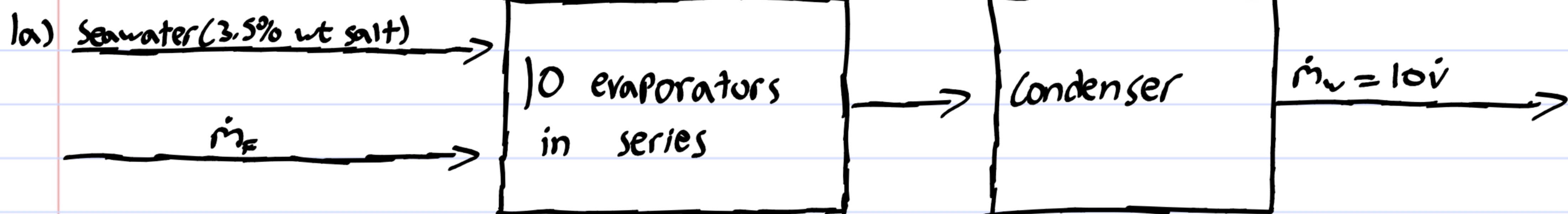


HW 4



1b) Salt in feed: $\dot{m}_s = .035 \dot{m}_F$

Salt in final brine: $\dot{m}_s = w_{b,s} \dot{m}_{10} = .05 \dot{m}_{10}$

Overall mass balance: $\dot{m}_F = \dot{m}_{10} + \dot{m}_w$

Equal vaporization per stage: $\dot{m}_w = 10\dot{v}$

Mass after y^{th} stage: $\dot{m}_y = \dot{m}_F - y\dot{v}$

Salt mass fraction after y^{th} stage: $w_{y,s} = \frac{\dot{m}_s}{\dot{m}_y}$

Fractional yield of fresh water: water in feed = $(1 - .035)\dot{m}_F = .965\dot{m}_F$, $Y = \frac{\dot{m}_w}{.965\dot{m}_F}$

1c) $\dot{m}_{10} = \frac{.035}{.05} \dot{m}_F = .70 \dot{m}_F$

$\dot{m}_w = \dot{m}_F - \dot{m}_{10} = .30 \dot{m}_F$

$\dot{v} = \frac{\dot{m}_w}{10} = .03 \dot{m}_F$

$\dot{m}_y = \dot{m}_F - y(.03 \dot{m}_F) = .98 \dot{m}_F$

$w_{y,s} = \frac{.035 \dot{m}_F}{.98 \dot{m}_F} = \frac{.035}{.98} = .03571 \rightarrow 3.57 \text{ wt \% salt}$

$Y = \frac{.30 \dot{m}_F}{.965 \dot{m}_F} = \frac{.30}{.965} = .311$

$Y = .311 \text{ kg H}_2\text{O recovered per kg H}_2\text{O in feed}$

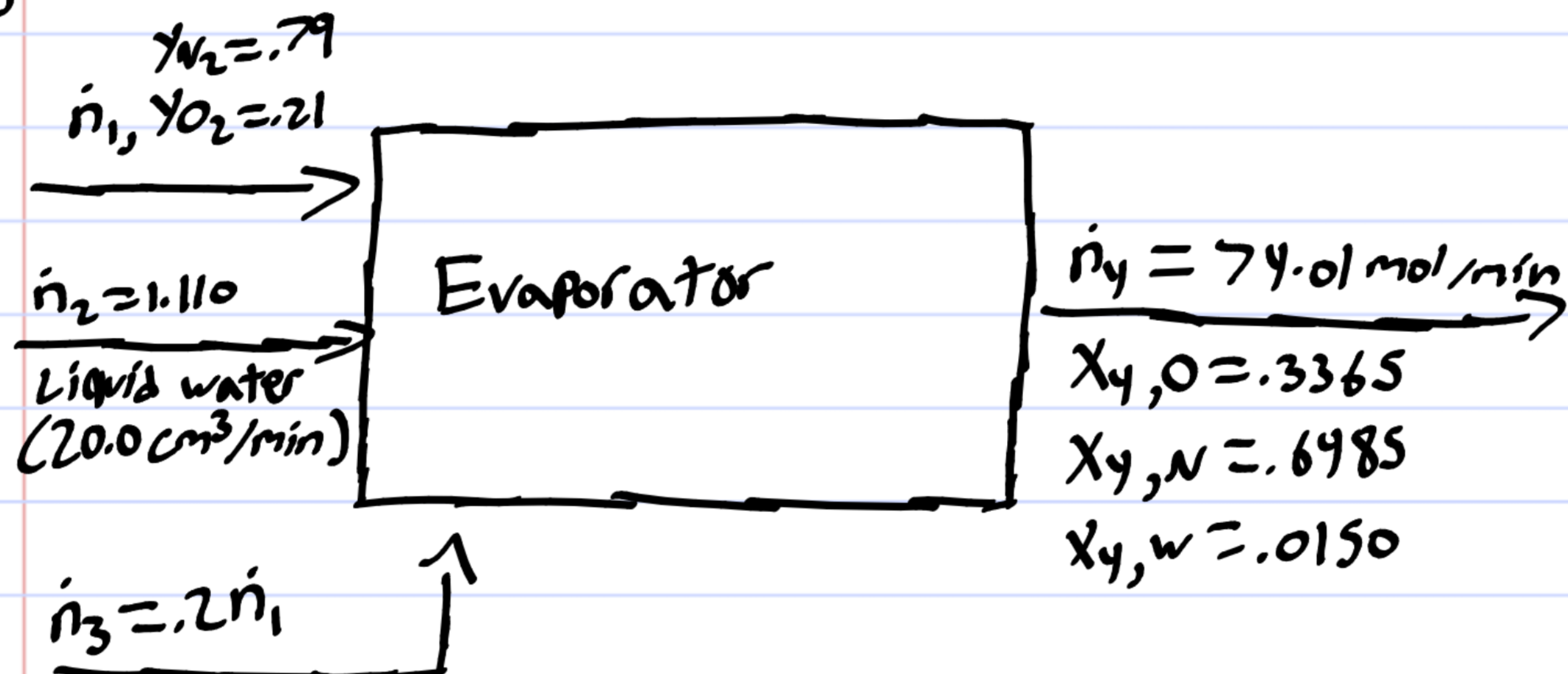
It does not need a basis because $\dot{m}_F$ cancels

1d) Brine flow after stage n : $\dot{m}_n = \dot{m}_F - n\dot{v}$

Salt is constant: $\dot{m}_s = .035 \dot{m}_F$ so $w_{n,s} = \frac{\dot{m}_s}{\dot{m}_n} = \frac{.035 \dot{m}_F}{\dot{m}_F - n\dot{v}}$, $\dot{v} = .03 \dot{m}_F$

$w_{n,s} = \frac{.035}{1 - .03n}$ $n = \text{stage so } n = 1, 2, \dots, 10$

2a)



2b) $\dot{n}_1, \dot{n}_3, \dot{n}_4, x_{4,O}, x_{4,N} = 5 \text{ unknowns}$

Known: $\dot{n}_2, x_{4,w} = .015$, air composition, $\dot{n}_3 = .2\dot{n}_1$

Equations: 1. $\dot{n}_3 = .2\dot{n}_1$

2. Total: $\dot{n}_4 = \dot{n}_1 + \dot{n}_2 + \dot{n}_3$

3. Water balance + given exit fraction: $\dot{n}_2 = x_{4,w} \dot{n}_4 = .015 \dot{n}_4$

4. O_2 balance: $x_{4,O} \dot{n}_4 = .21 \dot{n}_1 + \dot{n}_3$

5. N_2 balance: $x_{4,N} \dot{n}_4 = .79 \dot{n}_1$

DOF = 5 - 5 = 0

2c) $\dot{n}_2 = .015 \dot{n}_4$ $\dot{n}_4 = \frac{\dot{n}_2}{.015} = \frac{1.110}{.015} = 74.01 \text{ mol/min}$

$\dot{n}_4 = \dot{n}_1 + \dot{n}_2 + .2\dot{n}_1 = 1.2\dot{n}_1 + \dot{n}_2$

$\dot{n}_1 = \frac{\dot{n}_4 - \dot{n}_2}{1.2} = \frac{74.01 - 1.110}{1.2} = 60.75 \text{ mol/min}$

$\dot{n}_3 = .2\dot{n}_1 = 12.15 \text{ mol/min}$

$O_2 \text{ out moles} = .21 \dot{n}_1 + \dot{n}_3 = (.21 + .2)\dot{n}_1 = .41 \dot{n}_1$

$$X_{y,0} = \frac{.41\dot{n}_1}{\dot{n}_y} = \frac{.41(60.75)}{74.01} = .33654$$

$$N_2 \text{ out moles} = .79\dot{n}_1, \quad X_{y,N} = \frac{.79\dot{n}_1}{\dot{n}_y} = .64846$$

$$X_{y,w} = 1 - .3365 - .6485 = .0150$$

3) Raw pigment

4,000 lb/h (40 wt%)

2,000 lb/h (20 wt%)

4,000 lb/h (40 wt%)

Wash water

(6 lb washwater/lb pigment)

60,000 lb/h (pure water)



Washed pigment

8,000 lb/h (50 wt%)

$\leq .4$ lb/h ($\leq .005$ wt%)

3,999.6 lb/h (49.99 wt%)

waste water

62,000 lb/h

Salt 2,999.6 lb/h (3.23 wt%)

Water: 60,000 lb/h (96.77 wt%)

Feed: TiO_2 : $.4(10,000) = 4000$ lb/h

Salt: $.2(10,000) = 2000$ lb/h

Water: $.4(10,000) = 4000$ lb/h

Washwater optimum: $\dot{m}_w = 6 \frac{\text{lb water}}{\text{lb pigment}} (10,000) = 60,000$ lb/h

Total water into washer: $\dot{W}_in = 4000 + 60000 = 64,000$ lb/h, 100% of TiO_2 goes to washed pigment

Using only 2 constraints: washed pigment contains: $TiO_2 = 4000$, Salt S_p , water w_p

wastewater contains: Salt S_w , water w_w

Constraint 1: Salt limit in washed pigment

$$\frac{S_p}{4000} \leq \frac{100}{10^6} = 10^{-4} \quad S_p \leq 4000(10^{-4}) = .4 \text{ lb/h} \quad S_p \leq .4 \text{ lb/h}$$

Constraint 2: Optimum washwater feed rate

$\dot{W}_in = 64,000$ lb/h total water into washer

Salt balance: $S_p + S_w = 2000 \quad S_w = 2000 - S_p$

$S_w \geq 1999.6$ lb/h

Water balance: $w_p + w_w = 64,000$

- you cannot determine w_p and w_w from just those 2 constraints

- DOF: 4 unknowns: S_p, S_w, w_p, w_w

2 equations: salt and water balance

1 inequality and 1 given input

- So there is still 1 unresolved design choice, not fully specified yet

Using the remaining specs:

Additional spec 1: washed pigment is 50 wt% TiO_2

$$\frac{4000}{\dot{m}_{pig,out}} = .5 \quad \dot{m}_{pig,out} = 8000 \text{ lb/h}$$

$$4000 + S_p + w_p = 8000 \quad w_p + S_p = 4000$$

Additional spec 2: discharge regulation

discharge flow $\leq 30,000$ lb/h

Salt content $\leq .5$ wt% in discharge

$$S_p = .4$$

$$w_p = 4000 - .4 = 3999.6 \text{ lb/h}$$

$$S_w = 2000 - .4 = 1999.6 \text{ lb/h}$$

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$$W_w = 64000 - 3999.6 = 60,000.4 \text{ lb/h}$$

$$\dot{m}_{\text{waste}} = S_w + W_w = 1999.6 + 60,000.4 = 62,000 \text{ lb/h}$$

$$\text{washed pigment: } 9000 \text{ lb/h}$$

$$\text{waste water: } 62,000 \text{ lb/h}$$

discharge regulation:

flow limit: allowed $\leq 30,000 \text{ lb/h}$, actual wastewater = $62,000 \text{ lb/h}$

$62,000 > 30,000 \rightarrow$ violates flow limit

$$\text{Salt wt \% : } W_{\text{salt}} = \frac{1999.6}{62,000} = .03225 = 3.225\% \quad 3.225\% > .5\% \rightarrow \text{violates salt limit}$$

DOF analysis: The washer balances + product specs determine a unique wastewater stream. Then that unique wastewater stream cannot satisfy the discharge requirements, and so the overall process design is not feasible as specified

- a possible solution could be to not discharge to the river but rather send it to a facility to be neutralized

4) Fresh feeds: F_p = total fresh 95/5 propylene/propane stream

F_B = fresh benzene feed

Reactor inlet: $n_{2,p}$ propylene, $n_{2,I}$ propane (inert), $n_{2,B}$ benzene

Reactor extents: ξ_1 for $C_3H_6 + C_6H_6 \rightarrow C_9H_{12}$

ξ_2 for $C_3H_6 + C_9H_{12} \rightarrow C_{12}H_{18}$

Reactor outlet: $n_{3,p}$, $n_{3,I}$, $n_{3,B}$, $n_{3,C}$, $n_{3,D}$

Purge/recycle split of separator vapor: θ = purge fraction of separator vapor

$n_{7,p}$, $n_{7,I}$ = recycle vapor component flow rates back to mixer

$n_{6,p}$, $n_{6,I}$ = purge vapor component flow rates

17 unknown totals: F_p , F_B , $n_{2,p}$, $n_{2,I}$, $n_{2,B}$, ξ_1 , ξ_2 , $n_{3,p}$, $n_{3,I}$, $n_{3,B}$, $n_{3,C}$, $n_{3,D}$, θ , $n_{6,p}$, $n_{6,I}$, $n_{7,p}$, $n_{7,I}$

Equations used:

A) Fresh feed specification: $.95 F_p = 1.2 F_B$

B) propylene into reactor: $n_{2,p} = .95 F_p + n_{7,p}$

propane into reactor: $n_{2,I} = .05 F_p + n_{7,I}$

$$n_{2,B} = F_B + n_{3,B}$$

C) $n_{2,I} = .1 (n_{2,p} + n_{2,B} + n_{2,I})$

D) $n_{3,B} = n_{2,B} - \xi_1$

$$n_{3,p} = n_{2,p} - (\xi_1 + \xi_2)$$

$$n_{3,I} = n_{2,I}$$

$$n_{3,C} = \xi_1 - \xi_2$$

$$n_{3,D} = \xi_2$$

E) reactor conversion specs: $\xi_1 = .72 n_{2,B}$

$$\xi_1 + \xi_2 = .8 n_{2,p}$$

F) Separator + purge/recycle: $n_{6,p} = \theta n_{3,p}$ $n_{6,I} = \theta n_{3,I}$

$$n_{7,p} = (1 - \theta) n_{3,p} \quad n_{7,I} = (1 - \theta) n_{3,I}$$

G) Cumene production rate: $n_{3,C} = 25$

- 17 equations, 17 unknowns

$$\text{- DOF} = 17 - 17 = 0$$

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